Gravity and Air Resistance

Worked answer key: sketch, given, explained steps, final answer

Conventions used throughout

• **Notation:** v_1 = initial velocity, v_2 = final velocity, Δt = time, Δd = displacement, a = acceleration.

• **Signs:** each question says which direction is positive. In the working, every velocity, displacement and acceleration is just a number with a + or − sign. Only at the end is the sign turned back into a direction. For example, if west is positive, a = −1.36 m/s^2 means 1.36 m/s^2 [E].

• **g = 9.8 m/s^2** (the value that gives the answers printed on your worksheet). Full precision is kept in every step; only final answers are rounded.

EXAMPLE

Braking car: acceleration and stopping distance

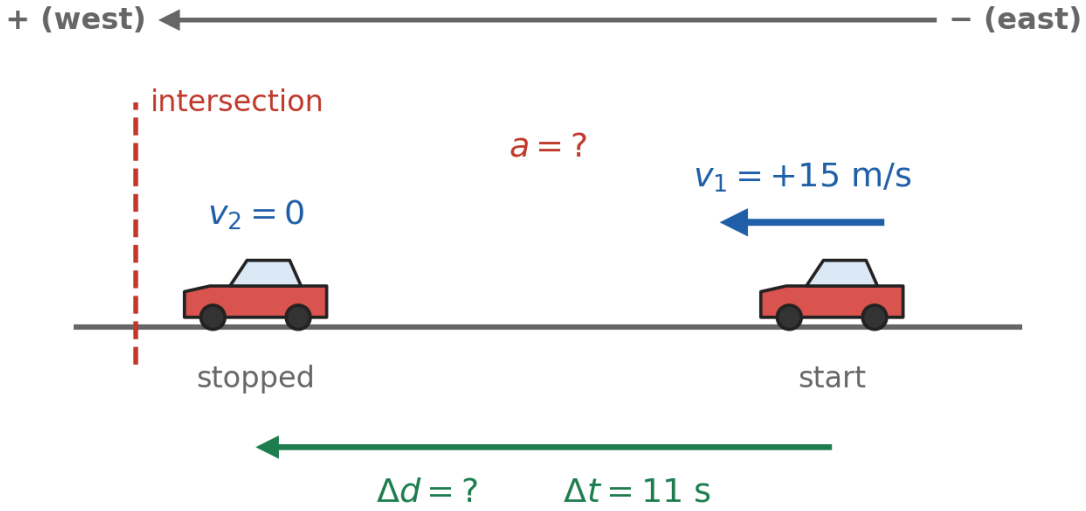
A car is approaching an intersection at 15 m/s [W]. It brakes and comes to a stop in 11s.

a) What was the car's acceleration?

b) What was the car's stopping distance?

Draw a sketch showing the car, indicating that west is positive, the initial velocity, the time, the final velocity and all variables you need to find:

The sketch: west is positive, and ? marks what we need to find



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$v_1 = +15$ m/s	$v_2 = 0$	$\Delta t = 11$ s
$a = ?$	$\Delta d = ?$	

Sign convention: west is positive and east is negative. The car moves west at 15 m/s, so $v_1 = +15$ m/s. It comes to a stop, so $v_2 = 0$.

Step 1 – (a) Acceleration

Acceleration is how quickly the velocity changes: the change in velocity divided by the time it took.

Formula: $a = \dfrac{v_2 - v_1}{\Delta t}$

$$a = \frac{0 - 15}{11}$$

$$a = \frac{-15}{11} = -1.36 \text{ m/s}^2$$

The answer is negative. West is positive, so negative means east: the acceleration is 1.36 m/s^2 [E]. It points the opposite way to the car's motion, which is what slowing down looks like.

Step 2 – (b) Stopping distance

The car slows down steadily, so its average velocity is halfway between its starting and ending velocities. Displacement = average velocity × time.

Formula: $\Delta d = \dfrac{v_1 + v_2}{2}\,\Delta t$

$$\Delta d = \frac{15 + 0}{2}\,(11)$$

$$\Delta d = (7.5)(11) = +82.5 \text{ m}$$

Positive means west: the car travels 82.5 m west before it stops. The stopping distance is how far it goes, 82.5 m.

(a) a = 1.4 m/s^2 [E]

(b) Stopping distance = 83 m

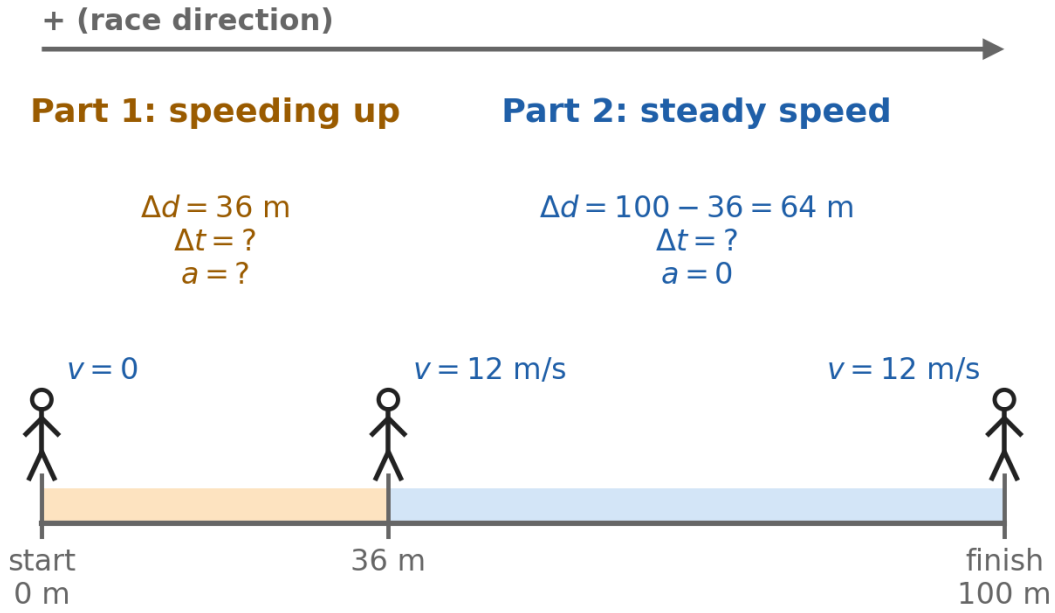
EXAMPLE

Sprinter: two parts of the race

A sprinter starts from rest in a 100 m race. She runs the first 36 m of the race at a constant acceleration reaching a speed of 12 m/s. She holds this speed until she finishes the race. What is her total time?

Sketch a diagram of the race, clearly indicating the two parts and labelling all the quantities you can.

The sketch: the two parts of the race (what we know so far)



Her motion changes at 36 m: first she speeds up, then she runs at a steady speed. So we split the race into two parts and solve each part on its own, using only that part's numbers. Then we add the two times.

Given, split by part. + is the race direction. She never turns around, so every value is positive.

	Part 1: start to 36 m (speeding up)	Part 2: 36 m to the finish (steady speed)
Speed at the start of the part, v_1	0 (she starts from rest)	12 m/s
Speed at the end of the part, v_2	12 m/s	12 m/s
Distance, Δd	36 m	100 − 36 = 64 m
Time, Δt	? (we call it $\Delta t_{\text{Part 1}}$)	? (we call it $\Delta t_{\text{Part 2}}$)
Acceleration, a	? (constant)	0 (her speed doesn't change)

Step 1 – Part 1: how long it takes

Use only the Part 1 column: $v_1 = 0$, $v_2 = 12$ m/s and $\Delta d = 36$ m. The acceleration is constant, so her average speed is halfway between 0 and 12 m/s, and distance = average speed × time.

Formula: $\Delta d = \frac{v_1 + v_2}{2}\Delta t$

Multiply both sides by 2: $2\,\Delta d = (v_1 + v_2)\,\Delta t$

Divide both sides by $(v_1 + v_2)$: $\Delta t = \frac{2\,\Delta d}{v_1 + v_2}$

Put in the Part 1 numbers: $\Delta t_{\text{Part 1}} = \frac{2(36)}{0 + 12} = \frac{72}{12} = 6.0 \text{ s}$

Step 2 – Part 1: her acceleration

The total time doesn't need this, but the sketch asks for every quantity we can find. Now that we know $\Delta t_{\text{Part 1}} = 6.0$ s:

Formula: $a = \frac{v_2 - v_1}{\Delta t}$

Put in the Part 1 numbers: $a_{\text{Part 1}} = \frac{12 - 0}{6.0} = +2.0 \text{ m/s}^2$

Positive means forward: she speeds up at 2.0 m/s^2.

Step 3 – Part 2: how long it takes

Use only the Part 2 column: she runs Δd = 64 m at a steady v = 12 m/s. With a steady speed, distance = speed × time.

Formula: $\Delta d = v\,\Delta t$

Divide both sides by v: $\Delta t = \frac{\Delta d}{v}$

Put in the Part 2 numbers: $\Delta t_{\text{Part 2}} = \frac{64}{12} = 5.33 \text{ s}$

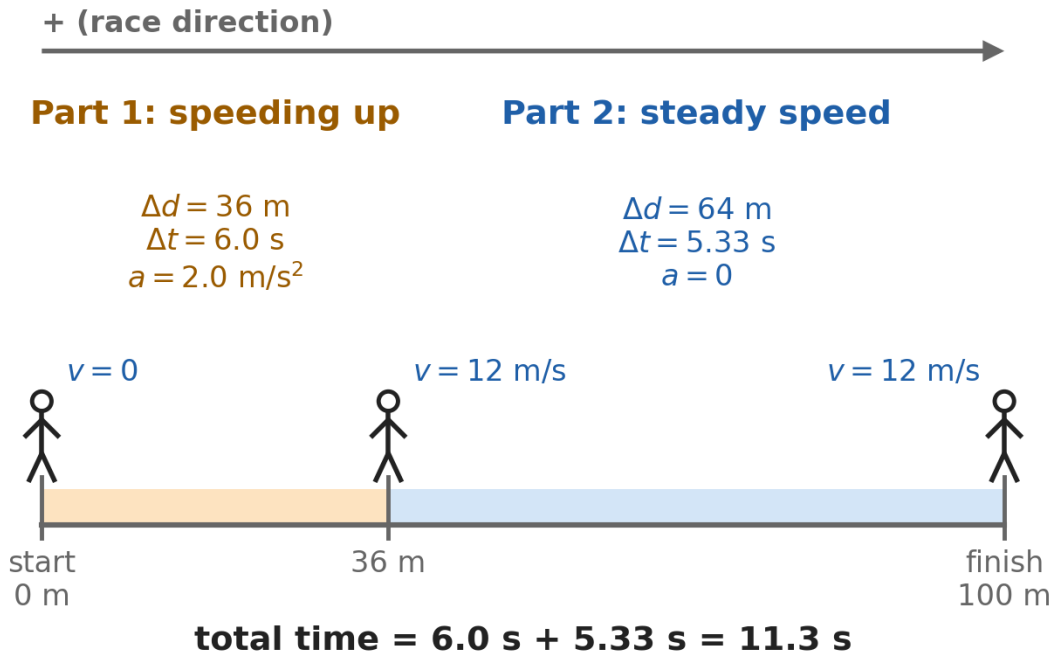
Step 4 – Total time

The two parts happen one after the other, so the total time is the two times added.

$\Delta t_{\text{total}} = \Delta t_{\text{Part 1}} + \Delta t_{\text{Part 2}}$

$\Delta t_{\text{total}} = 6.0 + 5.33 = 11.33 \text{ s}$

The completed sketch: every quantity labelled



total time = 6.0 s + 5.33 s = 11.3 s

Part 1 takes 6.0 s and Part 2 takes 5.3 s.

Total time = 11.3 s

QUESTION 1

Why gravity varies

> On or near the earth’s surface, the acceleration due to gravity is essentially a constant value of 9.8 m/s². It varies by a small amount depending on where you are on the earth. Explain why.

Three things change g slightly: latitude (in two ways), elevation and the rock underneath.

Factor	What is different	Effect on g
Latitude: Earth’s shape	Earth is slightly flattened at the poles, so a point at a pole is closer to Earth’s centre than a point on the equator.	The pull of gravity is stronger toward the poles.
Latitude: Earth’s spin	A point on the equator travels in the biggest circle as Earth spins, so part of the pull is used just to keep it moving in that circle. At the poles this effect is almost zero.	Measured g is smallest at the equator (about 9.78 m/s^2) and largest at the poles (about 9.83 m/s^2).
Elevation	Gravity gets weaker the farther you are from Earth’s centre.	g is slightly smaller on a mountain top than at sea level.
Rock underneath	Dense rock or ore underground adds extra pull; light material pulls less.	g varies slightly between places at the same latitude and elevation.

> **g varies with latitude (Earth’s shape and spin), elevation and the rock underneath. 9.8 m/s^2 is a convenient average for the surface.**

QUESTION 2

Ball thrown upward

A ball is thrown up into the air with an initial velocity of 16 m/s [up].

a) Sketch a diagram of the situation, labelling all known quantities. Clearly indicate which way is positive!

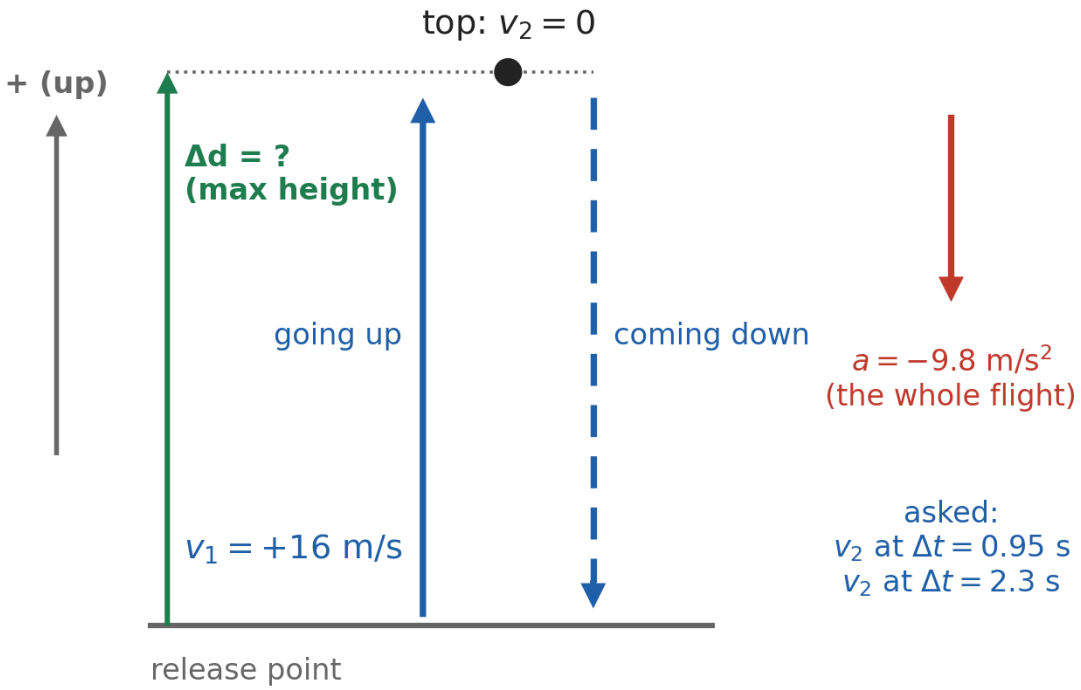
b) What is the velocity of the ball after

i) 0.95 s?

ii) 2.3 s?

c) What is the maximum height reached by the ball?

(a) The sketch: up is positive, and ? marks what we need to find



not to scale: the up and down paths are really the same vertical line

GIVEN

$v_1 = +16$ m/s $\qquad a = -9.8$ m/s^2 $\qquad v_2 = 0$ at the top

Sign convention: up is positive, so the upward throw is $v_1 = +16$ m/s and gravity (down) is $a = -9.8$ m/s^2. With no air resistance, gravity is the only force, so a stays the same for the whole flight: going up, at the top and coming down.

Step 1 – When does the ball reach the top?

At the top the ball stops for an instant, so $v_2 = 0$. Finding this time tells us whether 0.95 s and 2.3 s are on the way up or on the way down. Call this time Δt_{top}.

Formula: $v_2 = v_1 + a\,\Delta t$

Put in $v_2 = 0$: $0 = 16 + (-9.8)\,\Delta t_{top}$

Add $9.8\,\Delta t_{top}$ to both sides: $9.8\,\Delta t_{top} = 16$

Divide both sides by 9.8: $\Delta t_{top} = \frac{16}{9.8} = 1.63$ s

0.95 s is before 1.63 s, so the ball is still going up. 2.3 s is after 1.63 s, so it is coming down.

Step 2 – (b)(i) Velocity after 0.95 s

Use the same formula with Δt = 0.95 s.

Formula: $v_2 = v_1 + a\,\Delta t$

$v_2 = 16 + (-9.8)(0.95)$

$v_2 = 16 - 9.31 = +6.69$ m/s

Positive means up: 6.7 m/s [up]. The ball is still rising, but slower than when it was thrown.

Step 3 – (b)(ii) Velocity after 2.3 s

Same formula with Δt = 2.3 s.

Formula: $v_2 = v_1 + a\,\Delta t$

$v_2 = 16 + (-9.8)(2.3)$

$v_2 = 16 - 22.54 = -6.54$ m/s

Negative means down: 6.5 m/s [down]. This matches Step 1: by 2.3 s the ball is coming down.

Step 4 – (c) Maximum height

At the top $v_2 = 0$. We know v_1, v_2 and a, and we want Δd (the height from the release point to the top). The formula with exactly those four, and no time, is:

Formula: $v_2^2 = v_1^2 + 2a\,\Delta d$

Put in the numbers: $0 = 16^2 + 2(-9.8)\,\Delta d$

Work out each term: $0 = 256 - 19.6\,\Delta d$

Add $19.6\,\Delta d$ to both sides: $19.6\,\Delta d = 256$

Divide both sides by 19.6 : $\Delta d = \frac{256}{19.6} = +13.06$ m

Positive means up: the ball rises 13 m above the release point.

(b)(i) 6.7 m/s [up]

(b)(ii) 6.5 m/s [down]

(c) 13 m above the release point

QUESTION 3

What air resistance changes

> In the problem you solved above, you ignored air resistance. Consider what would change in your answers if you took air resistance into account. How would your values for velocity and maximum height change if we were to include air resistance? Why?

Air resistance always acts against the ball's motion, so it points down on the way up and up on the way down:

Stage	Forces on the ball	Acceleration	Effect on the answers
Going up	Gravity pulls down. Air resistance also pushes down (it always acts against the motion).	Down, and **bigger** than 9.8 m/s^2.	The ball slows down faster, so its velocity at 0.95 s is less than 6.7 m/s [up]. It reaches the top **sooner** and **lower** than 13 m.
Coming down	Gravity pulls down. Air resistance pushes up (against the motion).	Down, but **smaller** than 9.8 m/s^2.	It reaches the top earlier, so by 2.3 s it has been falling for longer, but it speeds up more slowly. It is moving down at 2.3 s, but its exact speed depends on how strong the air resistance is.

The exact new values can't be found from the information given. We would need the ball's mass, size and shape, and a formula for how air resistance depends on speed.

> **Going up, air resistance adds to gravity, so the ball slows faster: its velocity at 0.95 s is smaller and its maximum height is lower than 13 m. At 2.3 s it is moving down, but its exact speed depends on how strong the air resistance is.**

QUESTION 4

Why the constant-acceleration equations fail with air resistance

Air resistance becomes greater as you speed up. The force of air resistance on an object is proportional to the square of the objects speed. The faster you go, the more air resistance there is. $\mathbf{F_{air} \propto v^2}$

Based on this relationship, why would it be impossible for us to take air resistance into account when doing calculations with the equations given on page one?

Step 1 – What the page-one equations need

All of the page-one equations, such as $v_2 = v_1 + a\Delta t$, only work when the acceleration is **constant** (the same value the whole time).

Step 2 – The drag formula

Air resistance is also called drag. The drag force on an object moving through a fluid (air or water) is:

Formula: $F_{Drag} = \frac{1}{2} \rho A_c C v^2$

ρ = fluid density (the density of the air or water)
A_c = cross-sectional area (the size of the object's face pushing into the fluid)
C = drag coefficient (a number set by the object's shape)
v = speed of the object

For a given object in a given fluid, ½, ρ, A_c and C are all fixed numbers. Only v changes, so F_{Drag} is proportional to v^2, the same relationship as the worksheet's $F_{air} \propto v^2$.

Step 3 – What drag does to the acceleration

Think about a falling object (down is positive). Gravity pulls it down with its weight, mg, where m is its mass. Drag pushes up. The net force is weight minus drag, and the acceleration is net force divided by mass:

Formula: $a = \frac{F_{net}}{m}$

Put in $F_{net} = mg - F_{Drag}$: $a = \frac{mg - F_{Drag}}{m}$

Put in $F_{Drag} = \frac{1}{2}\rho A_c C v^2$: $a = \frac{mg - \frac{1}{2}\rho A_c C v^2}{m}$

Split the fraction: $a = \frac{mg}{m} - \frac{\frac{1}{2}\rho A_c C v^2}{m}$

Cancel m in the first term: $a = g - \frac{\frac{1}{2}\rho A_c C v^2}{m}$

Move the $\frac{1}{2}$ into the denominator: $a = g - \frac{\rho A_c C\, v^2}{2m}$

The acceleration depends on the speed v. As the object speeds up, v gets bigger, so the drag term gets bigger and the acceleration gets smaller. The acceleration keeps changing, so there is no single value of a to put into the page-one equations.

Drag depends on speed ($F_{Drag} = ½\rho A_c C v^2$), and speed keeps changing, so the acceleration keeps changing. The page-one equations need a constant acceleration, so they can't be used.

QUESTION 5

Terminal speed

> What is the term we use to describe an object’s motion when the force of air resistance is EQUAL to the force of gravity? (Hint: consider what would happen to the object’s speed at this point)

Step 1 – Balance the forces

For a falling object, gravity (its weight, mg) pulls down and drag pushes up. If the two forces are equal, they cancel:

Formula: $F_{net} = mg - F_{Drag}$

Put in $F_{Drag} = mg$: $F_{net} = mg - mg = 0$

Formula: $a = \frac{F_{net}}{m}$

Put in $F_{net} = 0$: $a = \frac{0}{m} = 0$

With zero acceleration, the speed stops changing. The object doesn’t stop: it keeps falling at the same steady speed.

Step 2 – What the terminal speed depends on

Use the drag formula to find the speed v at which F_{Drag} = mg. Call it v_t:

Formula: $F_{Drag} = \frac{1}{2}\rho A_c C v^2$

Set $F_{Drag} = mg$ and $v = v_t$: $\frac{1}{2}\rho A_c C v_t^2 = mg$

Multiply both sides by 2: $\rho A_c C v_t^2 = 2mg$

Divide both sides by $\rho A_c C$: $v_t^2 = \frac{2mg}{\rho A_c C}$

Take the square root of both sides: $v_t = \sqrt{\frac{2mg}{\rho A_c C}}$

A heavier object (bigger m) has a higher terminal speed. A larger cross-sectional area A_c, like an open parachute, gives a lower terminal speed.

Terminal velocity (terminal speed): the steady speed an object falls at when drag equals the force of gravity. Its value is $v_t = \sqrt{(2mg / \rho A_c C)}$.